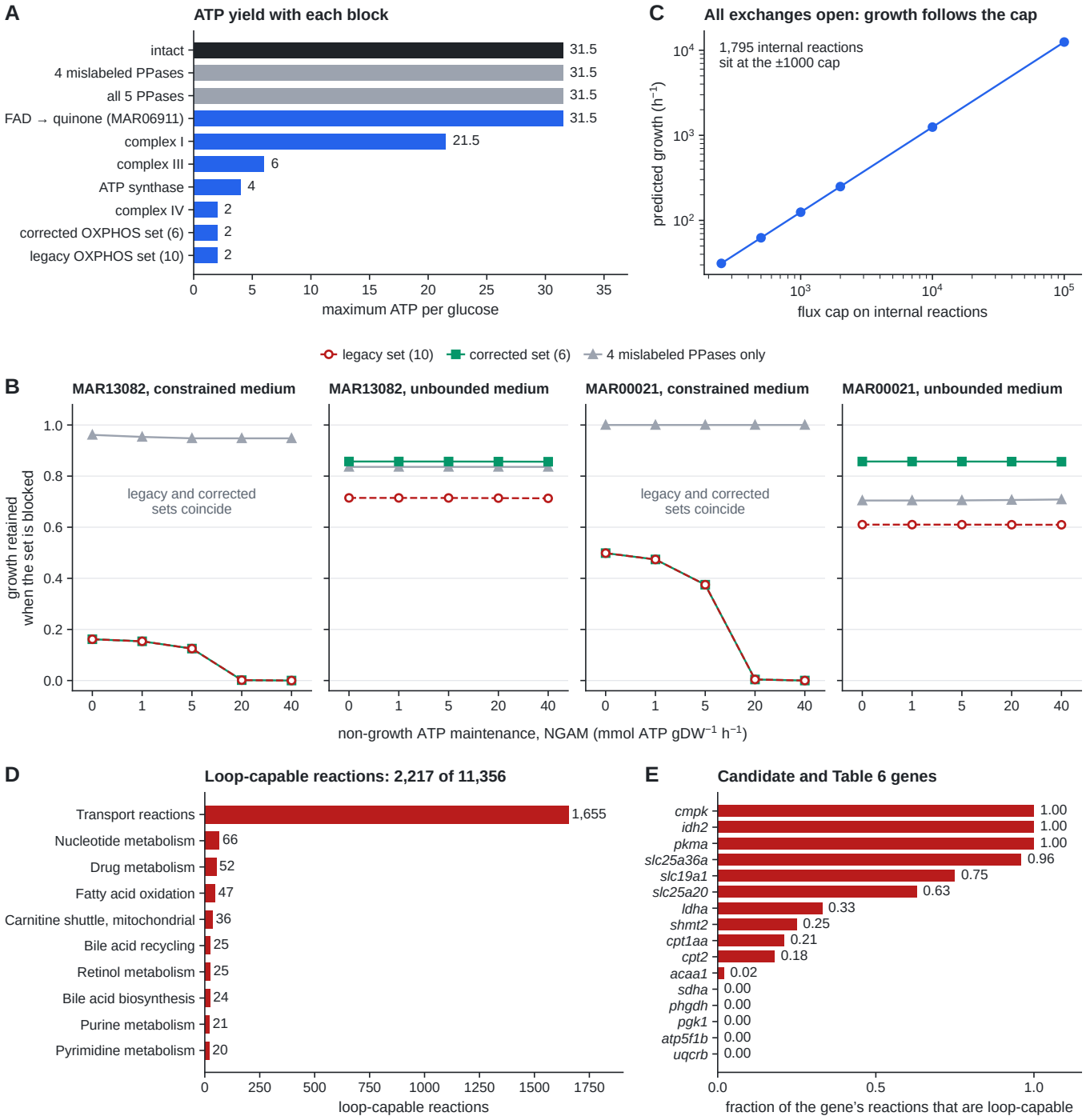
A
ATP yield with each block
intact
4 mislabeled PPases
all 5 PPases
FAD → quinone (MAR06911)
complex I
complex III
ATP synthase
complex IV
corrected OXPHOS set (6)
legacy OXPHOS set (10)
31.5
31.5
31.5
31.5
21.5
6
4
2
2
2
0 5 10 15 20 25 30 35
maximum ATP per glucose
C
All exchanges open: growth follows the cap
1,795 internal reactions
sit at the ±1000 cap
predicted growth (h⁻¹)
10² 10³ 10⁴
10³ 10⁴ 10⁵
flux cap on internal reactions
legacy set (10)
corrected set (6)
4 mislabeled PPases only
B
MAR13082, constrained medium
MAR13082, unbounded medium
MAR00021, constrained medium
MAR00021, unbounded medium
growth retained when the set is blocked
0.0 0.2 0.4 0.6 0.8 1.0
legacy and corrected sets coincide
legacy and corrected sets coincide
0 1 5 20 40
non-growth ATP maintenance, NGAM (mmol ATP gDW⁻¹ h⁻¹)
D
Loop-capable reactions: 2,217 of 11,356
Transport reactions
Nucleotide metabolism
Drug metabolism
Fatty acid oxidation
Carnitine shuttle, mitochondrial
Bile acid recycling
Retinol metabolism
Bile acid biosynthesis
Purine metabolism
Pyrimidine metabolism
1,655
66
52
47
36
25
25
24
21
20
0 250 500 750 1000 1250 1500 1750
loop-capable reactions
E
Candidate and Table 6 genes
cmpk
idh2
pkma
slc25a36a
slc19a1
slc25a20
ldha
shmt2
cpt1aa
cpt2
acaa1
sdha
phgdh
pgk1
atp5f1b
uqcrb
1.00
1.00
1.00
0.96
0.75
0.63
0.33
0.25
0.21
0.18
0.02
0.00
0.00
0.00
0.00
0.00
0.0 0.5 1.0
fraction of the gene's reactions that are loop-capable